

Example 2.1: Complex Powers

Complete step-by-step solutions for simplifying exponents of the imaginary unit i

SUBDIVISION (I): SIMPLIFY i^7

CONCEPT Division Algorithm

Any integer exponent of the imaginary unit repeats its value in a cycle of 4. We divide the exponent 7 by 4 to extract the quotient and find the remaining cyclic factor:

$$7 = 4 (1) + 3$$

Thus, the value reduces to the remainder index power, i^3 .

SOLUTION Algebraic Steps

Simplifying the exponent index:

$$\begin{aligned} i^7 &= i^{4 (1) + 3} \\ &= (i^4)^1 \cdot i^3 \\ &= 1 \cdot (- i) = - i \end{aligned}$$

SUBDIVISION (II): SIMPLIFY i^{1729}

CONCEPT Ramanujan Number Index

We divide the large index 1729 by 4 to extract the quotient and find the remaining cyclic factor:

$$1729 = 4 (432) + 1$$

Since $(i^4)^{432} = 1^{432} = 1$, the index simplifies to a basic power of 1.

SOLUTION Mathematical Steps

Simplifying the exponent index:

$$\begin{aligned} i^{1729} &= i^{1728 + 1} \\ &= i^{4 (432)} \cdot i^1 \\ &= (i^4)^{432} \cdot i = 1 \cdot i = i \end{aligned}$$

SUBDIVISION (III): DUAL POWER

PART 1 Evaluate i^{-1924}

Since -1924 is a perfect multiple of 4:

$$-1924 = 4 (-481)$$

$$i^{-1924} = (i^4)^{-481} = 1^{-481} = 1$$

PART 2 Evaluate i^{2018} & Sum

Since 2018 divided by 4 leaves a remainder of 2:

$$2018 = 4 (504) + 2$$

$$i^{2018} = i^2 = -1$$

Adding both parts: $1 + (-1) = 0$.

SUBDIVISION (IV): SIGMA SUM

CONCEPT Symmetrical Quad Rule

The sum of any four consecutive powers of the imaginary unit i is always zero:

$$i^k + i^{k+1} + i^{k+2} + i^{k+3} = 0$$

Since the index goes up to 102, we have 25 complete cycles of 4 terms ($25 \times 4 = 100$), leaving the final 2 terms.

SOLUTION Series Expansion

Writing out the expansion and simplifying:

$$\begin{aligned}\sum_{n=1}^{102} i^n &= 0 + i^{101} + i^{102} \\ &= i^1 + i^2 \\ &= i - 1 = -1 + i\end{aligned}$$

SUBDIVISION (v): PRODUCT SEQUENCE

STEP 1 Summing the Exponents

Applying the exponent addition rule for multiplied bases:

$$i^1 \cdot i^2 \cdot i^3 \cdots i^{40} = i^{1+2+3+\cdots+40}$$

We use the natural summation formula $\frac{n(n+1)}{2}$:

$$\text{Sum} = \frac{40 \cdot 41}{2} = 820$$

STEP 2 Base Simplification

Reducing the consolidated power i^{820} using the divisibility of 4:

$$\begin{aligned} i^{820} &= i^{4(205)} \\ &= (i^4)^{205} \\ &= 1^{205} = 1 \end{aligned}$$